final_report

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0.1 Final Report: Voting Demographics

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0.1.1 Introduction

This project explores how the demographics of voters affect their participation in elections, using voter turnout data from the 2024 U.S. presidential election. The policy problem we address is unequal voter participation across demographic groups, which can undermine democratic representation.

Understanding which demographic groups are more or less likely to vote is important for policymakers and election officials seeking to increase overall voter turnout. If certain groups consistently participate at lower rates, their preferences may be underrepresented in electoral outcomes.

Using the results of our analysis, we aim to predict whether an individual voter is likely to vote. Such predictions could help inform targeted outreach strategies designed to encourage participation among voters who are less likely to vote. In this context, identifying individuals who ultimately do not vote is particularly important, as policy interventions would be directed toward these groups.

In a democratic society, voter turnout is a critical indicator of political participation, as higher turnout allows a broader range of citizens' voices to be reflected in electoral outcomes.

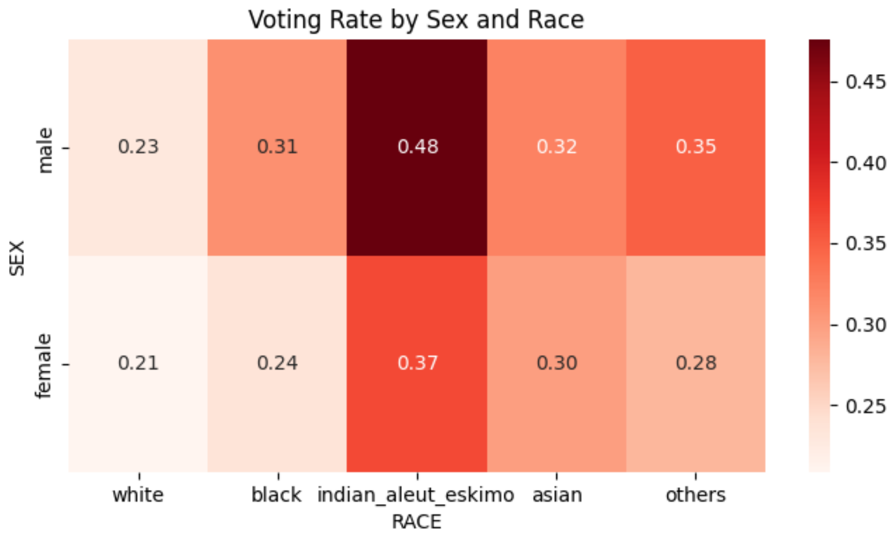
To make these predictions, we employ three models: logistic regression, random forests, and neural networks. We evaluate their performance using the F1 score, which is appropriate for our analysis because the dataset is imbalanced, as discussed in the Data section. We define not_voted as 1 and voted as 0, as our analysis focuses on identifying individuals who do not vote.

0.1.2 Data

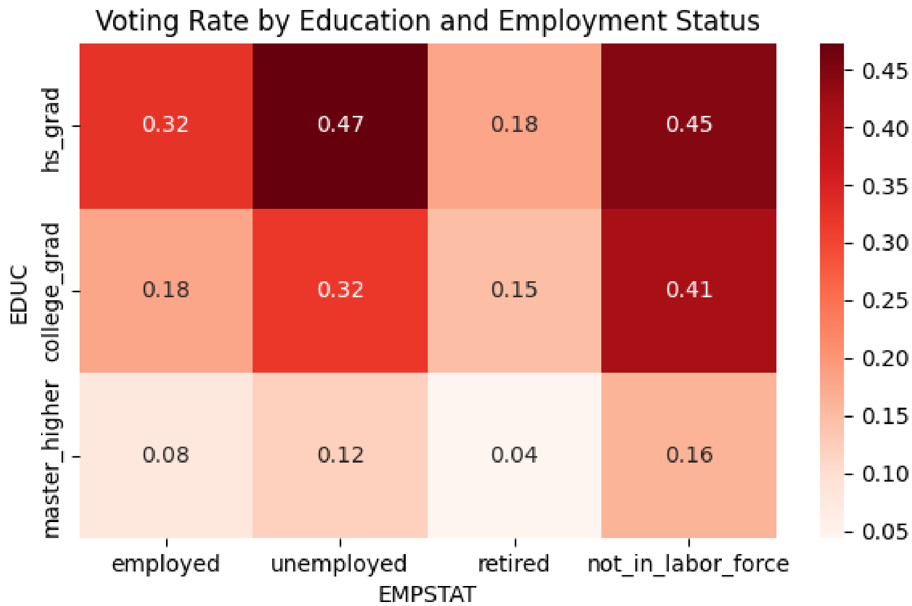
Our data source is the IPUMS CPS (Current Population Survey) November 2024 Voting Supplement (https://cps.ipums.org/). The dataset includes information on whether individuals voted in the 2024 U.S. presidential election, along with a wide range of demographic characteristics. From the available variables, we selected 14 features based on their expected relationship with voting behavior:

- Sex (categorical): Sex is a standard demographic factor that may influence voting behavior. We included it because many previous studies suggest differences in political participation across sexes. In our dataset, males exhibit a slightly higher non-voting rate than females.
- Race (categorical, manipulated): Race is another important demographic variable affecting voting behavior. We grouped all smaller or mixed racial categories into an "Other" category

to reduce sparsity, as these categories contained relatively few observations. Our data show that individuals identified as American Indian or Eskimo have the highest non-voting rates.

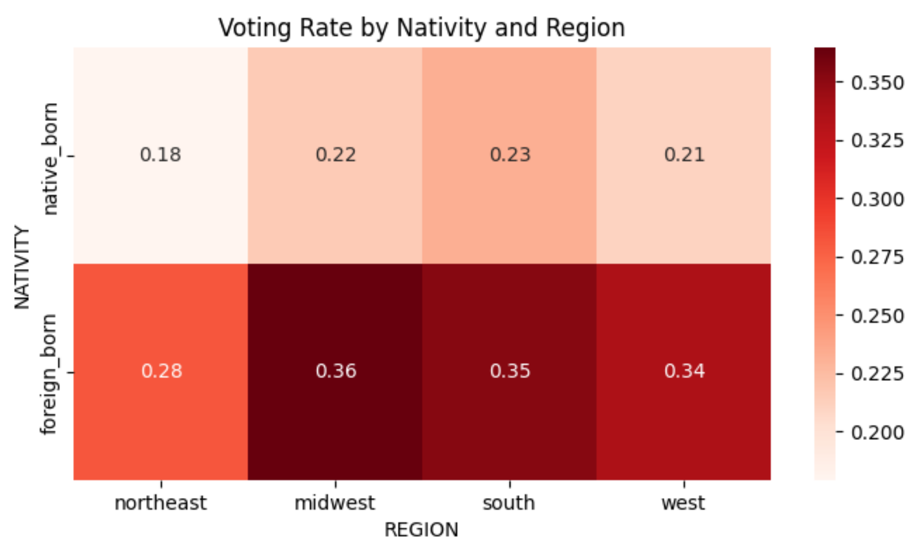


- Education (categorical, manipulated): Education level can influence political engagement and interest. We initially categorized education into four groups: below high school, high school graduate, college graduate, and master's degree or higher. However, after data cleaning, no individuals remained in the below high school category. Individuals who attended college but did not graduate were grouped with high school graduates. As a result, all individuals in the final dataset had at least a high school education. Our results show that high school graduates have the highest non-voting rates.
- Employment Status (categorical, manipulated): Employment status may affect voting behavior through differences in time availability and political engagement. We simplified the original categories into four groups: employed, unemployed, retired, and not in the labor force. Our analysis shows that unemployed and not-in-labor-force individuals tend to have higher non-voting rates.



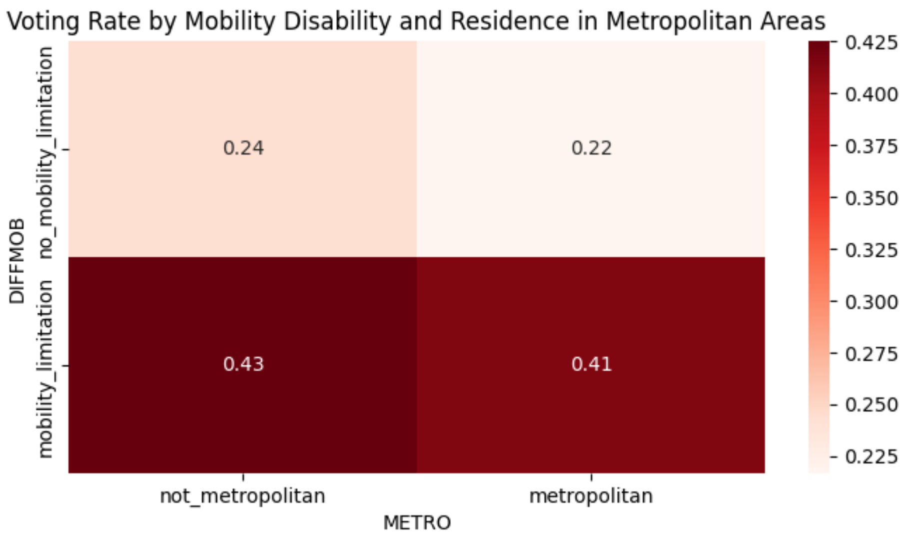
- Nativity (categorical, manipulated): Nativity indicates whether an individual is native-born or foreign-born. We simplified the variable by grouping individuals with at least one native-born parent as "native-born," and others as "foreign-born." This reflects differences in familiarity and engagement with U.S. political systems. Foreign-born individuals appear to have higher non-voting rates.

- Region (categorical, manipulated): Political behavior often varies by geographic region. We aggregated subregions into broader categories (Northeast, Midwest, South, and West). The Midwest and South show slightly higher non-voting rates, though differences are relatively modest.

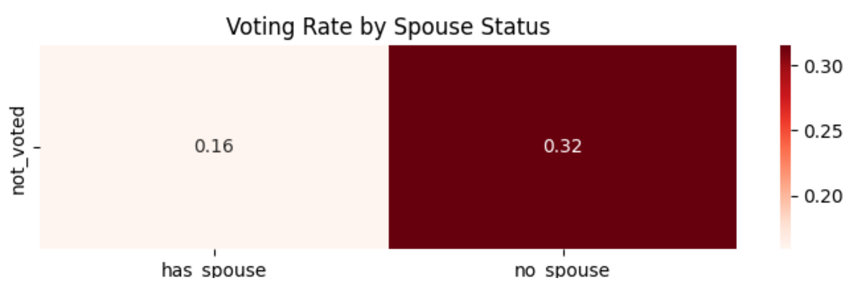


- Mobility Disability (categorical): Mobility limitations may restrict physical access to polling locations. Individuals with mobility-related disabilities show higher non-voting rates in our data.

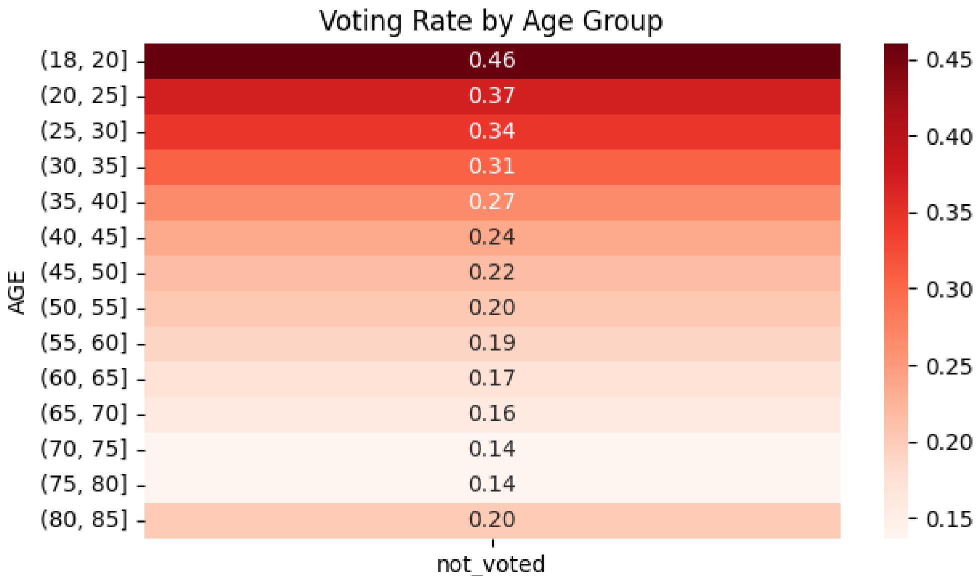
- Metropolitan (categorical, manipulated): Access to polling locations may differ between metropolitan and non-metropolitan areas. We reduced this variable to a binary indicator (metropolitan vs. non-metropolitan). Individuals living in non-metropolitan areas tend to have slightly higher non-voting rates.



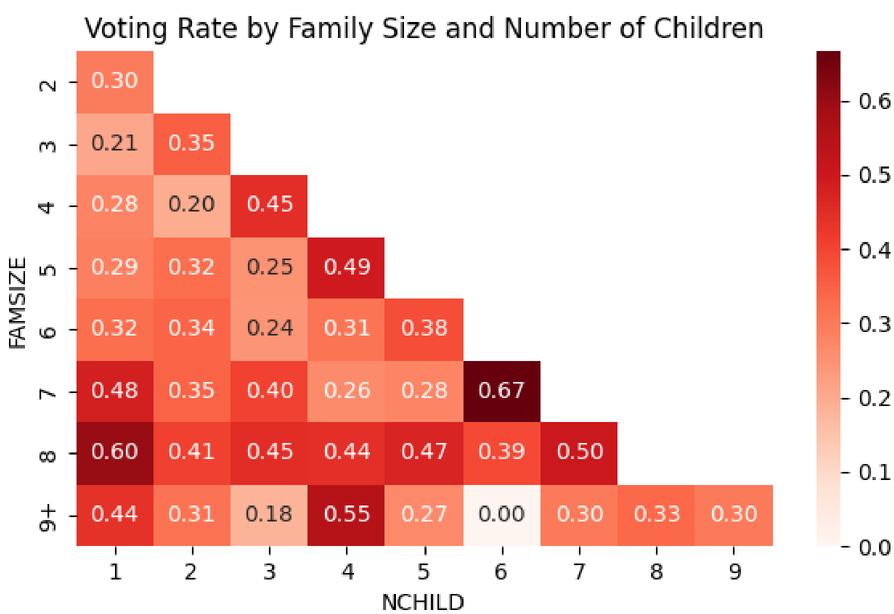
- Marital Status (categorical, manipulated): Marital status may influence voting through social interaction and shared decision-making. We simplified this variable into two categories: with spouse and without spouse. Individuals without a spouse tend to have higher non-voting rates.



- Age (numerical, manipulated): Age is a key predictor of voting behavior. In the dataset, individuals aged 80–84 are recorded as 80, and those aged 85+ are recorded as 85. To address this limitation, we grouped age into five-year intervals and used the midpoint of each interval. Younger individuals show significantly higher non-voting rates.

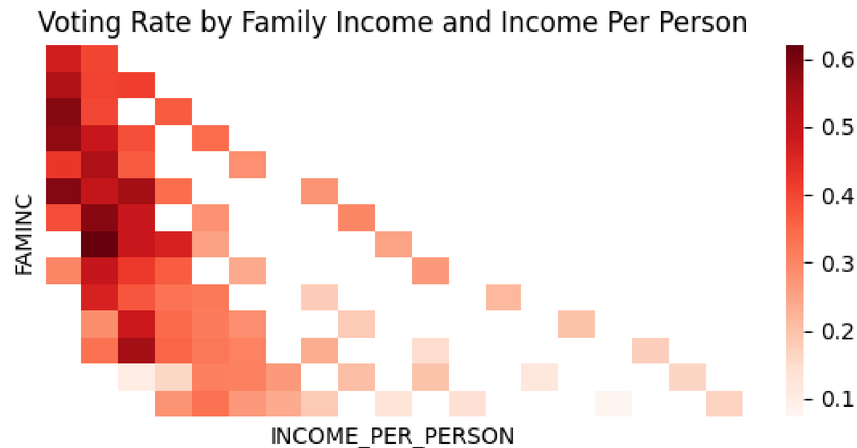


- Family Size (numerical): Larger family size may reduce available time for civic participation. Our data suggest that non-voting rates generally increase with family size.
- Number of Children (numerical): Having children may limit time available for voting. Our dataset indicates that households with more children tend to have higher non-voting rates.



- Family Income (numerical, manipulated): Income can influence political participation. Since income was provided in ranges rather than exact values, we used the midpoint of each range as an estimate. Individuals from lower-income households tend to have higher non-voting rates.

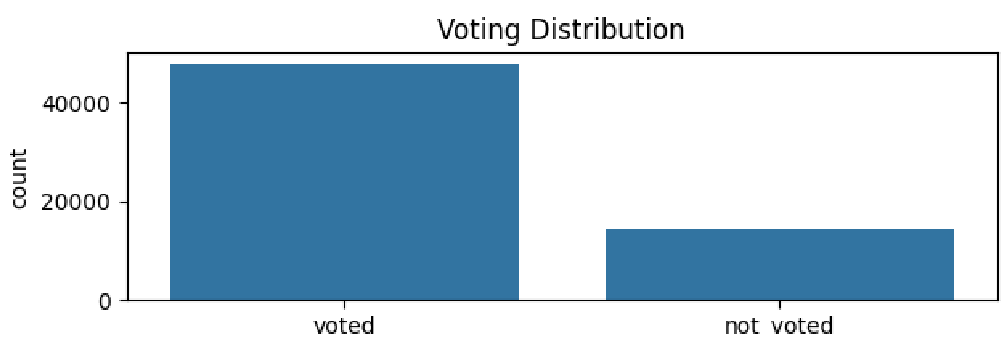
- Income Per Person (numerical, extracted): We created this feature by dividing family income by family size, as family income alone does not account for household size. This provides a more accurate measure of economic resources per individual. Lower income per person is associated with higher non-voting rates.



We cleaned the dataset by removing observations with missing values. If more than one feature was missing (marked as "not in universe"), the entire row was dropped. After cleaning, the dataset contained 62,327 observations.

We then performed feature engineering and transformations as described above, including grouping categories, creating new variables, and simplifying complex variables. After that, we encoded categorical variables into numerical representations for model input using OneHotEncoder, and normalized numerical variables where appropriate using StandardScaler. Finally, we split the data into training (70%), validation (10%), and test (20%) sets for model development.

One major limitation of the data is class imbalance, as there are significantly more individuals who voted than those who did not. This imbalance affects both model training and evaluation. We observed consistently high accuracy across different hyperparameter settings, making accuracy a less informative metric. Therefore, we used the F1 score for validation and evaluation, as it better captures the balance between precision and recall in imbalanced classification problems.



Additionally, several variables are reported in categories rather than exact values. For example:

- Family income is provided in ranges rather than precise values.

- Age is top-coded (80 and 85+).

These limitations reduce precision of our measurements.

We would have liked to include variables related to transportation access, such as:

- Car ownership
- Access to public transportation

These factors could significantly affect physical access to polling locations. In the absence of such variables, we attempted to proxy access using metropolitan status and mobility disability.

0.1.3 Model 1: Logistic Regression

How did you pick your models and tune them? What was the best-performing model out of all you tried, given the metrics you used? What was the worst-performing model? Any ideas why?

0.1.4 Model 2: Random Forests

How did you pick your models and tune them? What was the best-performing model out of all you tried, given the metrics you used? What was the worst-performing model? Any ideas why?

0.1.5 Model 3: Neural Networks

Neural networks are capable of learning complex, nonlinear relationships in data. Our goal is to predict whether an individual voted based on demographic characteristics, and neural networks are well-suited for capturing such patterns.

To train the model, we used the PyTorch library. The training process consists of the following steps: (1) for each training batch*, perform a forward pass to compute predictions, (2) calculate the loss by comparing the predicted values (y_hat) with the true labels (y), (3) update model parameters using backpropagation, and (4) repeat this process over multiple epochs.

Because the dataset is imbalanced, we applied class weighting to give more importance to the minority class (not_voted). Specifically, we used the following formulation:

[ ]:
```
self.pos_weight = torch.tensor([(self.torch_dataset.y_train == 0).sum()/(self.
↪torch_dataset.y_train == 1).sum()]).float()
```

This increases the penalty for misclassifying individuals who did not vote, which aligns with the focus of our analysis.

We also observed that some models were unstable. Models with high variability in validation F1 scores (standard deviation greater than 0.1) were excluded to ensure stable performance:

[ ]:
```
if f1_std > 0.1:
    result["f1"] = 0
    results.append(result)
    continue
```

Moreover, we used a stability-adjusted score (mean F1 minus standard deviation) to select the best model, ensuring both high performance and consistent behavior across training epochs..

[ ]:
```
score = current_f1_mean - f1_std
```

In addition, models were excluded if their training loss failed to decrease or increased after several epochs, indicating poor convergence:

[ ]:
```
elif epoch > 5 and avg_loss > initial_loss:
    return []
```

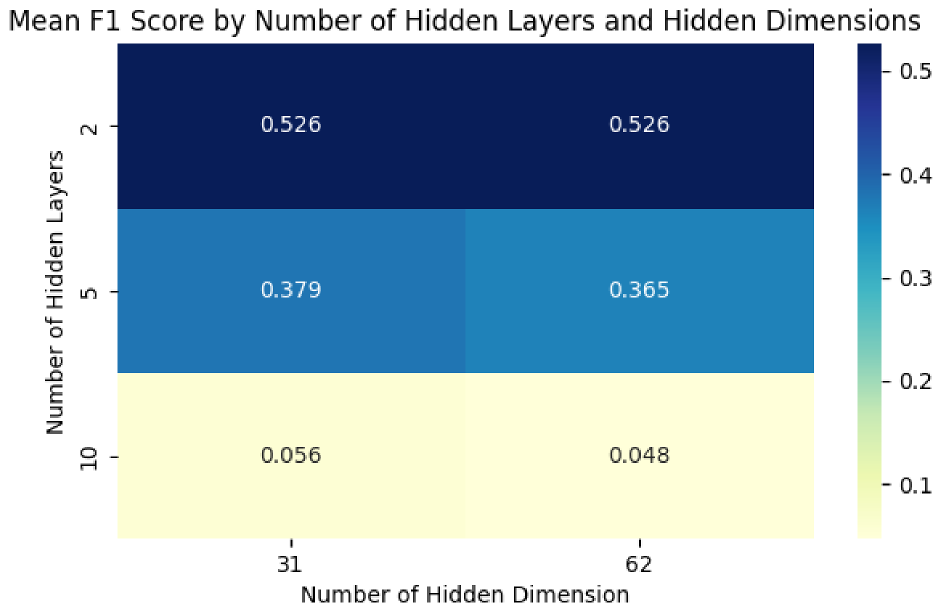
We tuned five key hyperparameters:

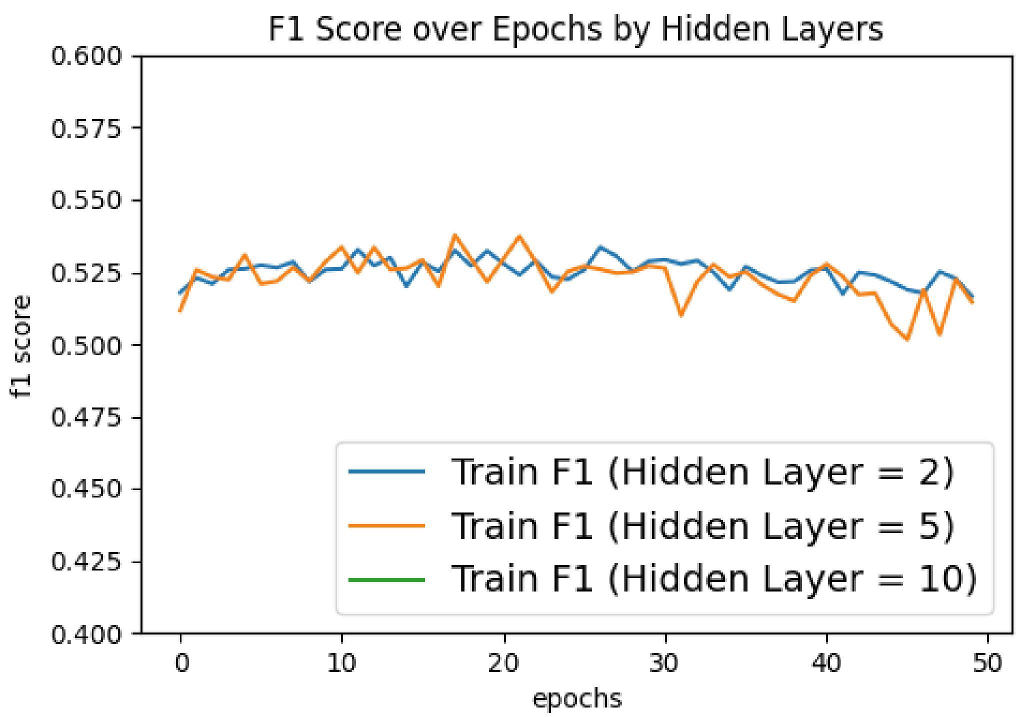
- Batch Size: To improve training efficiency compared to pure stochastic gradient descent (SGD), we used mini-batch training with batch sizes of 50, 100, and 200. For validation and testing, predictions were made without batching.
- Number of Hidden Layers: This determines the depth of the network. We tested models with 2, 5, and 10 hidden layers. While deeper networks can capture more complex patterns, they are also more prone to overfitting and instability.
- Hidden Dimension: This refers to the number of neurons in each hidden layer. Since the input dimension was 31 (after encoding), we tested 31 and 62. Larger dimensions increase model complexity.
- Learning Rate: This controls how quickly the model updates its parameters. A learning rate that is too high may cause divergence, while a rate that is too low may slow learning. We tested 0.01, 0.001, and 0.0001.
- Penalty Term: This represents the strength of L2 regularization. Strong regularization may lead to underfitting, while weak regularization may cause overfitting. We tested values of 0.01, 0.001, and 0.0001.

After running the hyperparameter search, we identified the best-performing model based on the validation F1 score. Hyperparameter selection was based on the validation set rather than the training set.

- Best Model: Batch size = 50, Number of hidden layers = 5, Hidden dimension = 62, Learning rate = 0.001, Penalty rate = 0.0001

However, as can be seen on the heatmap below, we found that models with a smaller number of hidden layers generally performed better, suggesting that simpler architectures were sufficient for this task and helped reduce overfitting.





The model with 5 hidden layers achieved the highest score due to occasional performance peaks. However, models with 2 hidden layers showed more consistent performance across epochs, indicating greater stability.

The training curves show that performance quickly stabilizes and fluctuates slightly across epochs, suggesting limited improvement from additional training and reinforcing the importance of stability in model selection.

Deeper architectures showed signs of overfitting, which further supports the choice of a simpler model. (Hidden layer with 10 is not shown in the graph, as these models were highly unstable and were excluded during training.)

Using the best model, we evaluated performance on the test set. The number of false positives (predicting not_voted when the individual actually voted) and false negatives (predicting voted when the individual actually did not vote) were relatively similar in magnitude, indicating that the model achieved relatively balanced performance across classes.

- F1: 0.5
- False Negative Rate: 0.37767795438838975
- False Positive Rate: 0.262118679481822

Despite the flexibility of neural networks, the relatively modest F1 score suggests that the model may not be well-suited to this dataset. This may be due to the size of the dataset, class imbalance, or the tabular nature of the data, where simpler models often perform more effectively. Compared to logistic regression and random forest models, the neural network did not provide a meaningful improvement in performance, indicating that increased model complexity does not necessarily lead to better predictive results in this case.

0.1.6 Surprises

Did anything unexpected happen? What parts of the project were surprisingly difficult?

0.1.7 Conclusion

What did you learn about your research question? What did you learn about machine learning through the tasks? How would you continue this research if you wanted to further investigate? If you had more time and infinite resources, how would you continue your project?